

Revision Notes for Class 8 Maths Chapter 2 Power Play

Power Play introduces the fascinating concept of exponents and exponential growth through simple, relatable examples and activities. The chapter begins with an activity where students try folding a sheet of paper repeatedly, quickly revealing the surprising nature of exponential increase. With each fold, the thickness of the paper doubles, and soon, after only a few folds, the thickness becomes unexpectedly large. For instance, after 10 folds, a 0.001 cm-thick paper becomes slightly over 1 cm thick, and after 30 folds, it can reach about 10.7 km—almost the height at which airplanes fly.

Exponential Growth Explained

Exponential, or multiplicative, growth means that a quantity increases by a fixed multiple at each step. In the case of the folded paper, the thickness doubles with each fold. This is a great way to understand powers of numbers. After every 10 folds, the thickness increases by 1024 times because it's multiplied by 2^{10} . The chapter uses clear tables to highlight this doubling pattern, with thickness values at folds 1, 2, 3, and so on, showing how quickly the numbers grow.

Exponential Notation and Operations

Exponential notation is a shorthand to represent repeated multiplication. For example, $n \times n = n^2$ ("n squared"), $n \times n \times n = n^3$ ("n cubed"), and so on. The number being multiplied (n) is called the base, and the number of times it is multiplied is the exponent or power. So, $2^{10} = 1024$. This notation helps express very large (or small) numbers easily.

- $p^4 \times p^6 = p^{10}$
- $n^a \times n^b = n^{a+b}$
- $n^a \div n^b = n^{a-b}$

Powers of 10 and Scientific Notation

Many large and small numbers in daily life and science are written using powers of 10. For example, 1000 can be written as 10^3 , and 0.01 as 10^{-2} . The expanded form of a number such as 47561 is written as $(4 \times 10^4) + (7 \times 10^3) + (5 \times 10^2) + (6 \times 10^1) + (1 \times 10^0)$. Scientific notation helps to write very large or small numbers efficiently. For instance, 5900 is written as 5.9×10^3 , and 8 million becomes 8×10^6 in scientific notation.

Operations with Exponents

The chapter shows how to multiply and divide numbers with exponents. For example, $n^a \times n^b = n^{a+b}$ and $(n^a)^b = n^{a \times b}$. Negative exponents represent reciprocals: $n^{-a} = 1/n^a$. Any non-zero number raised to the power zero is 1, i.e., $n^0 = 1$. These identities let us solve and simplify exponential expressions quickly.

Exponential Growth vs. Linear Growth

Exponential growth is different from linear (additive) growth. For example, to reach the moon by walking 20 cm steps, it would take over 192 crore steps—a linear process. But using exponential growth, such as doubling with each step (like paper folds), much fewer steps are needed to reach similar distances. This contrast highlights how powerful exponential changes can be in real life.

Applying Exponents to Real Life

The concept of combinations is related to exponents. If you have 4 dresses and 3 caps, you can create $4 \times 3 = 12$ unique combinations. In digital security, a 5-digit password has $10^5 = 100,000$ possible arrangements. Understanding these concepts helps students calculate possibilities and analyze situations effectively.

Large Numbers and Units

The chapter names large numbers in both the Indian and International systems. For example:

lakh: 10^5 (1,00,000)

crore: 10^7 (1,00,00,000)

million: 10^6 (1,000,000)

billion: 10^9 (1,000,000,000)

Tables in the chapter give a feel for the size of numbers in seconds, population estimates of rare species, and other real-world quantities. For example, 10^6 seconds is about 11.57 days and 10^9 seconds is roughly 31.7 years.

Estimation and Thinking with Exponents

Examples like Tulābhāra (weighing a person against goods) show practical uses of multiplication, estimation, and large numbers. Students are encouraged to estimate and guess before calculating, building both intuition and numeracy.

Sample Questions and Practice

Express $6 \times 6 \times 6 \times 6$ in exponential form.

Find the value of $2^{100} \div 2^{25}$.

Simplify $(13^{-2})^{-3}$.

Express 64^3 as a product of powers in three ways.

Check if cube numbers are always square numbers.

Summary and Quick Facts

Exponents (powers) are used to write repeated multiplication in short form.

Exponential identities: $n^a \times n^b = n^{a+b}$, $(n^a)^b = n^{ab}$, $n^a \div n^b = n^{a-b}$, $n^0 = 1$ for $n \neq 0$,
 $n^{-a} = 1/n^a$.

Scientific notation helps express very large or small numbers as $x \times 10^y$.

Linear growth is additive while exponential growth is multiplicative.

Estimation and understanding patterns in exponential growth are useful skills for maths and science.

